Solar Water Heater



Category: **Solar, Power & Energy** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Solar water heater

What it is used for

A solar water heater uses solar collectors to heat water for domestic or other approved applications. Operate it within the system's pressure and temperature limits and follow safe procedures for hot water, plumbing, and electrical components.

Main parts

- Solar collector panels or evacuated tubes
- Insulated storage tank
- Water inlet, outlet, and circulation piping
- Temperature and pressure relief valve
- Mounting frame and insulation
- Controller, pump, sensors, and electrical connections, if fitted

Before you start

1. Identify the system type and read the manufacturer's operating instructions.
2. Inspect collectors, tank, piping, insulation, valves, and mounting structure for damage or leakage.
3. Check that the water supply and circulation path are available and unobstructed.
4. Confirm that pressure relief devices and isolation valves are correctly installed and accessible.
5. For pumped systems, inspect the pump, controller, sensors, and electrical connections.
6. Keep the service area accessible and use suitable PPE when inspecting hot components.

How to use

1. Ensure the system is correctly filled and vented according to its design before operation.
2. Check the water supply and open the designated valves in the specified sequence.
3. For active systems, switch on the controller and circulation pump as instructed.
4. Verify that temperature and pressure readings remain within the system's rated limits.
5. Inspect visible pipe joints and connections for leaks without touching hot surfaces.
6. Allow the system to heat water under normal operating conditions.
7. Monitor controller indications and check for abnormal noise, leakage, or pressure changes.
8. Use hot water cautiously and test its temperature before contact or use.
9. If a fault or overpressure warning occurs, follow the system's shutdown and service procedure.
10. Isolate the water or electrical supply only when safe and according to the instructions.
11. Allow components to cool before cleaning, inspection, or maintenance.
12. Record faults and arrange qualified service for leaks, relief-valve issues, or control failures.

⚠ Safety

- Installation and repair should be performed by qualified plumbing or electrical personnel as applicable.
- Water from the tank and outlet may be hot enough to cause scalds.
- Never block or disable the temperature and pressure relief valve.
- Do not open pressurised pipes or the tank while the system is hot.
- Use safe access equipment when working on roof-mounted collectors.
- Disconnect electrical power before servicing pumps or controllers and follow isolation procedures.

Care & storage

- Inspect collectors, tank insulation, pipes, supports, and valves at scheduled intervals.
- Clean collector surfaces using the manufacturer's approved method.
- Check for leaks, corrosion, scale buildup, and damaged insulation.
- Test or service relief valves according to the manufacturer's instructions and local requirements.
- Arrange qualified servicing for reduced heating performance, abnormal pressure, or recurring faults.

Common mistakes to avoid

- Operating the system without the required water fill or circulation.
- Ignoring leaks, corrosion, or damaged insulation.
- Blocking the pressure relief valve or exceeding rated limits.
- Touching hot pipes or working on roof collectors without safe access and protection.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.